Sami Mostfa

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Profile

Professional Master's in Computer Engineering student at Polytechnique Montréal, seeking an internship in Montréal for the **Summer 2027** term (May 3 – August 31, 2027), primarily in **Data Engineering**, **MLOps** or **ML Engineering** — while remaining open to other opportunities.

My dual background in software engineering (**ENSEEIHT**) and AI/ML (**Polytechnique Montréal**) has given me a solid foundation in both model design/training and infrastructure deployment. During my internship at *Axians HPC*, I contributed to the design of a **Kubernetes** architecture for integrating an **LLM** inference server. I am now looking to put this combined skill set to work on an end-to-end project — from design to production — as an **ML Engineer** or **MLOps Engineer**.

Education

Polytechnique Montréal 08/2025 – 08/2027

Professional Master's in Computer Engineering - Montréal

- Current cumulative GPA: **3.83/4.0**
- Coursework: Artificial Intelligence, Machine Learning, Continuous Optimization, Data Mining, Data Visualization, Bayesian Statistics, Natural Language Processing, Intelligent DevOps for Large-Scale Software Systems, Cybersecurity

ENSEEIHT 09/2023 – 06/2025

Software Engineering - Toulouse

- Coursework: Software & Systems Engineering, Concurrent & Communicating Systems, Databases, Graph Theory, Automata Theory, Computability & Complexity, Computer Architecture, Object-Oriented Technology, Networks, Formal Methods, Semantics & Language Translation

Preparatory Classes (CPGE) — Lycée Claude Fauriel 09/2021 – 07/2023

MPI / MP2I - Saint-Étienne*

- MPI*: Mathematics, Physics, Computer Science
- MP2I: Mathematics, Physics, Computer Science, Engineering Sciences

Experience

High-Performance Computing Engineering Intern 06/2025 – 08/2025

Axians HPC (High Performance Computing) *Labège, France*

- **Solution put into production by Axians in 2026**
- Designed an architecture integrating **Kubernetes** with the **Open OnDemand** (OOD) HPC portal, working around the native limitations of the OOD adapter (inability to natively deploy *Deployments*)
- Developed a *kubectl* wrapper in **Bash** that intercepts adapter calls and redirects *get/delete* commands to the actual resources, making job status display on the web interface reliable
- Explored **Kyverno** policies for automatic cleanup of orphaned resources, quota management and access hardening (NFS mounts, GPU) on the cluster
- Deployed and configured an **LLM (vLLM)** inference server on the Kubernetes cluster
- Wrote a technical architecture document formalizing the OOD workflows (**Apache**, **NGINX**, **Passenger** and Kubernetes adapter) for the client and the team
- Hard Skills: **Linux Systems**, **Kubernetes**, **Bash**

Private Tutor in Mathematics & Computer Science ↗ 09/2020 – 06/2025

Superprof — Self-employed *Toulouse, France (Hybrid)*

- Taught private mathematics and computer science lessons alongside my studies to approximately **15 high school students**, for a total of about **80 hours**
- Soft Skills: **Communication**, **Adaptability** to each student, **Autonomy** in managing an activity over several years

Lifeguard 07/2024 – 09/2024

Escap'ad Aquatic Center *Saint Laurent de Chamousset, France*

- Active supervision of the pools to ensure swimmer safety & emergency response in case of accidents
- Soft Skills: **Sense of responsibility**, **Decision-making** in critical situations

Projects

Graph Clustering via Modularity Maximization | *Julia* ↗ 07/2026 – Present

- Implemented a research paper by Professor Daniel Aloise, instructor of **Data Mining** (INF8111) at Polytechnique Montréal
- Optimized modularity using several algorithms: **LPA**, **LPAm**, **LPAm+** (classic greedy and multi-step strategies) on 9 benchmark graphs from the DIMACS Implementation Challenge (from 34 to over 31,000 vertices)
- **Reviewed 5 scientific papers** to ground the algorithm implementations and validate results
- Real-world applications: identifying affinity groups in **social networks**, financial **fraud detection**, better understanding of protein interactions in **biology**
- Hard Skills: **Data Science**, **Graph Theory**, **Applied Mathematics**

Montréal Air Quality Prediction for 2025 | *Julia* ↗ 01/2026 – 04/2026

- **Team project (5 people)** that earned a grade of **95/100** in the course **Probabilistic and Statistical Methods Applied to AI** (MTH3302), Winter 2026 term
- Predicted fine particulate matter at **7 stations** in Montréal from **17 years** of historical data, totaling about **33,000 samples**
- Pipeline: variable-specific handling of missing data (temporal interpolation, median imputation, removal), followed by engineering of more than **20 features** (lags, moving averages, weather interactions, seasonal and wildfire indicators)
- Compared 4 alternative approaches: temporal weighting, per-station models, regime splitting, **VIF/AIC** selection
- Selected the best model by **RMSE** minimization and submitted it on **Kaggle**
- Hard Skills: **Time Series**, **Linear Regression**, **Feature Engineering**

Basketball Detection and Localization with Deep Learning | *Python* 03/2025 – 05/2025

- **Team project (4 people)** for the Deep Learning course (N9EN19C) at ENSEEIHT
- Designed a **multi-output CNN** (**4 convolutional blocks**, 32 to 256 filters) combining **presence classification** and **bounding-box regression** to localize the ball
- Implemented **from scratch** the **IoU** (Intersection over Union) metric, vectorized with **TensorFlow** tensors, along with **Precision/Recall/F1-score** at a 40% overlap threshold
- Multi-task training over **25 epochs** (combined **BinaryCrossentropy + MSE** loss, **Adam** optimizer), with image normalization (**64x64px**) and standardization of bounding-box coordinates
- Real-world applications: **automated sports video analysis**, **real-time object tracking**
- Hard Skills: **Deep Learning**, **Computer Vision (CNN)**, **TensorFlow/Keras**

Technical Skills

Languages: Python, Julia, C, Java, TypeScript, JavaScript, OCaml, Ada, Bash, HTML5, CSS3, LaTeX
Data Science: NumPy, Pandas, Scikit-Learn, Seaborn, XGBoost, DataFrames.jl, Distributions.jl, Statistics.jl
Data Visualization: D3.js, Matplotlib, Gadfly.jl, OpenGL
Machine Learning, AI & Deep Learning: PyTorch, TensorFlow, Keras, GLM.jl
Environments: VS Code, Vim, GCP, Google Colab, Eclipse, Jupyter Notebook
DevOps & Infrastructure: Kubernetes, Docker, NGINX, Apache, Kyverno, Ruby
Agentic AI: OpenRouter, MCP (Model Context Protocol), Claude Code, Prompt Engineering, Custom Skills & Tools Development
Data Formats: SQL, CSV, JSON, Parquet, XML, YAML, XLSX, Markdown, TeX
Languages (spoken): French (Fluent), English (B2 – Cambridge Linguaskill Certification), Spanish (A2), Arabic (Basic)

Interests

Music: **14 years** of guitar practice & advanced knowledge of music theory — youtube.com/@sami.musiic ↗
First Aid & Rescue: **400 hours** of cumulative training — National Lifeguard & Aquatic Rescue Certificate (8 months of training) & Team First Aid level 1 (40h of training)
Guitar Lessons: **10 hours** taught, in addition to private math and computer science tutoring
Sports: Swimming & Weightlifting